

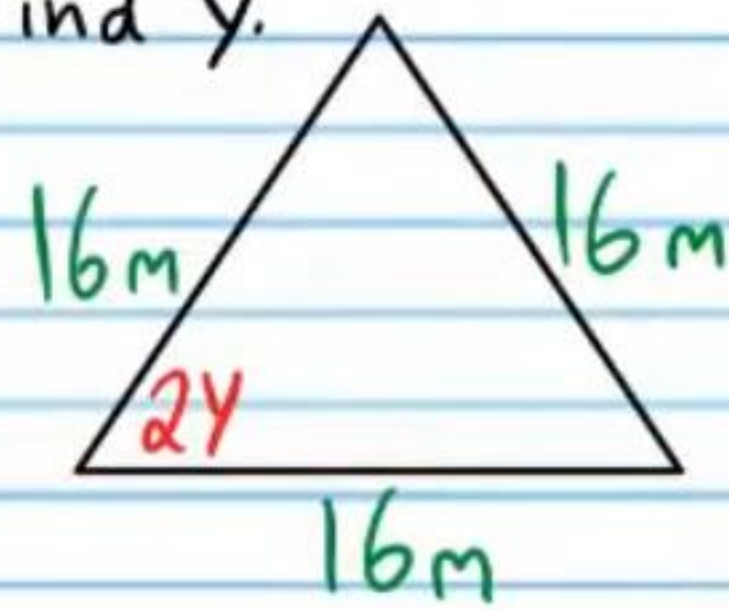
Practice Geometry Test II (Version 1)

Name
Date
Course

Instructions

- I) Write your name, the date, and the course title (i.e. Geometry).
- II) Show work when necessary.
- III) If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- IV) Each problem is worth four points for a total of 100 points.

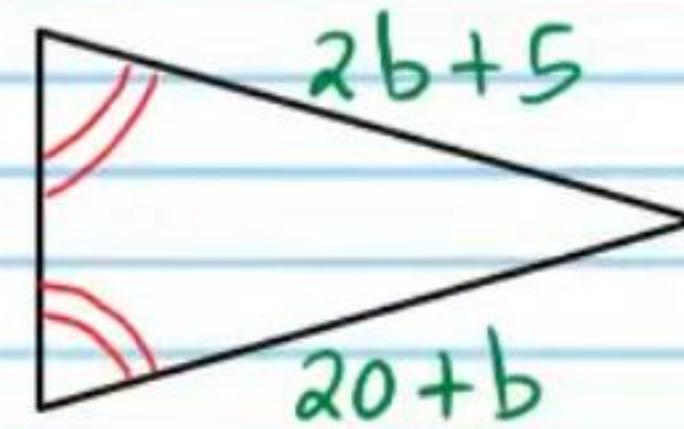
① Find y .



$$2y = 60^\circ$$

$$\boxed{y = 30}$$

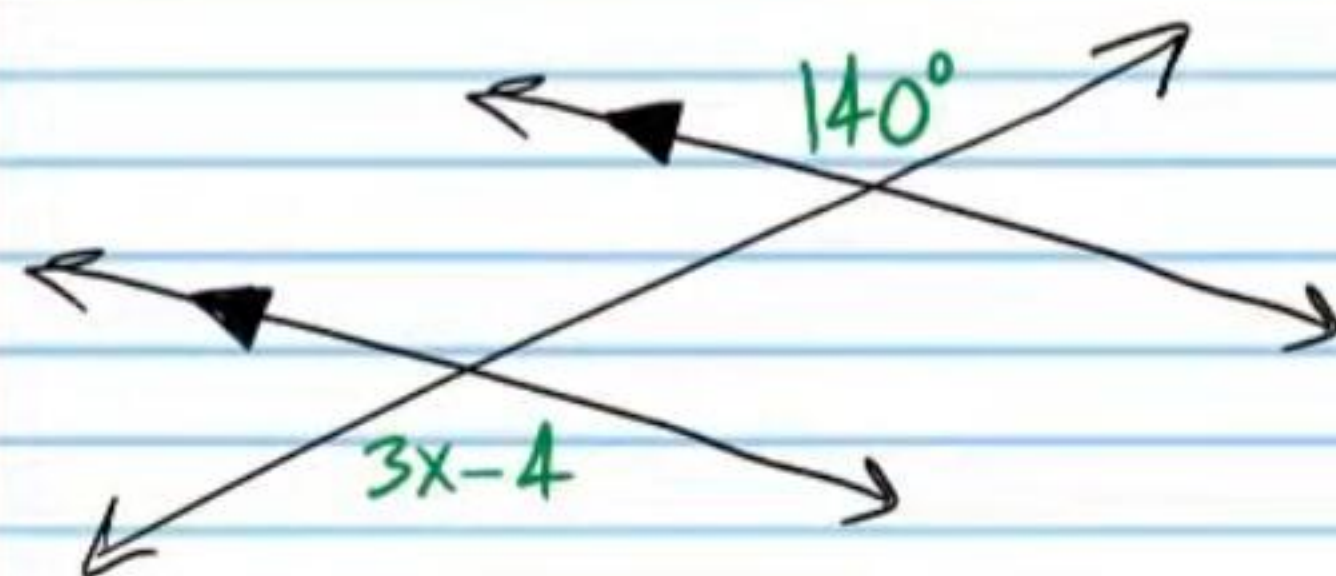
② Find b .



$$2b + 5 = 20 + b$$

$$\boxed{b = 15}$$

③ Find x and state the theorem or postulate used. You do not have to write a two-column proof but you must show the equation used to find x .



AE AT

$$140 = 3x - 4$$

$$144 = 3x$$

$$\boxed{48 = x}$$

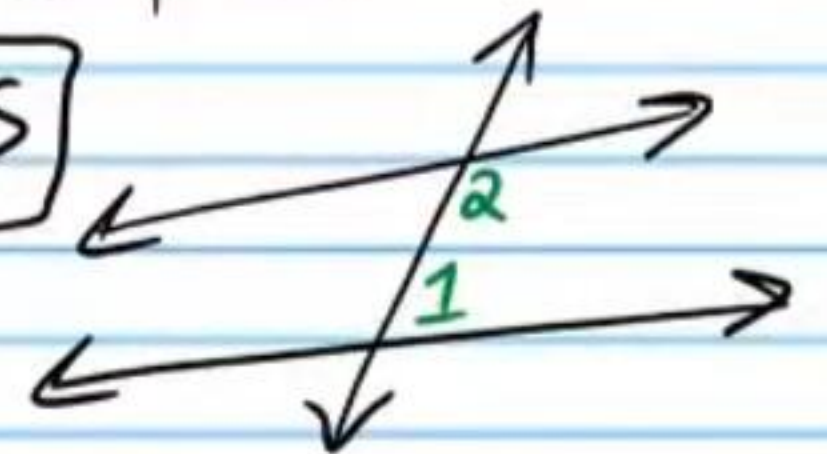
④ i) Is the following statement true or false?

The Triangle Sum Theorem can only be used with right triangles.

False

ii) Name the following angle pair.

Consecutive Interior Angles



⑤ If the tire shop is halfway between the dealership and the gas station, what are the coordinates of the gas station?

$$M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right) = M(-5, 3)$$

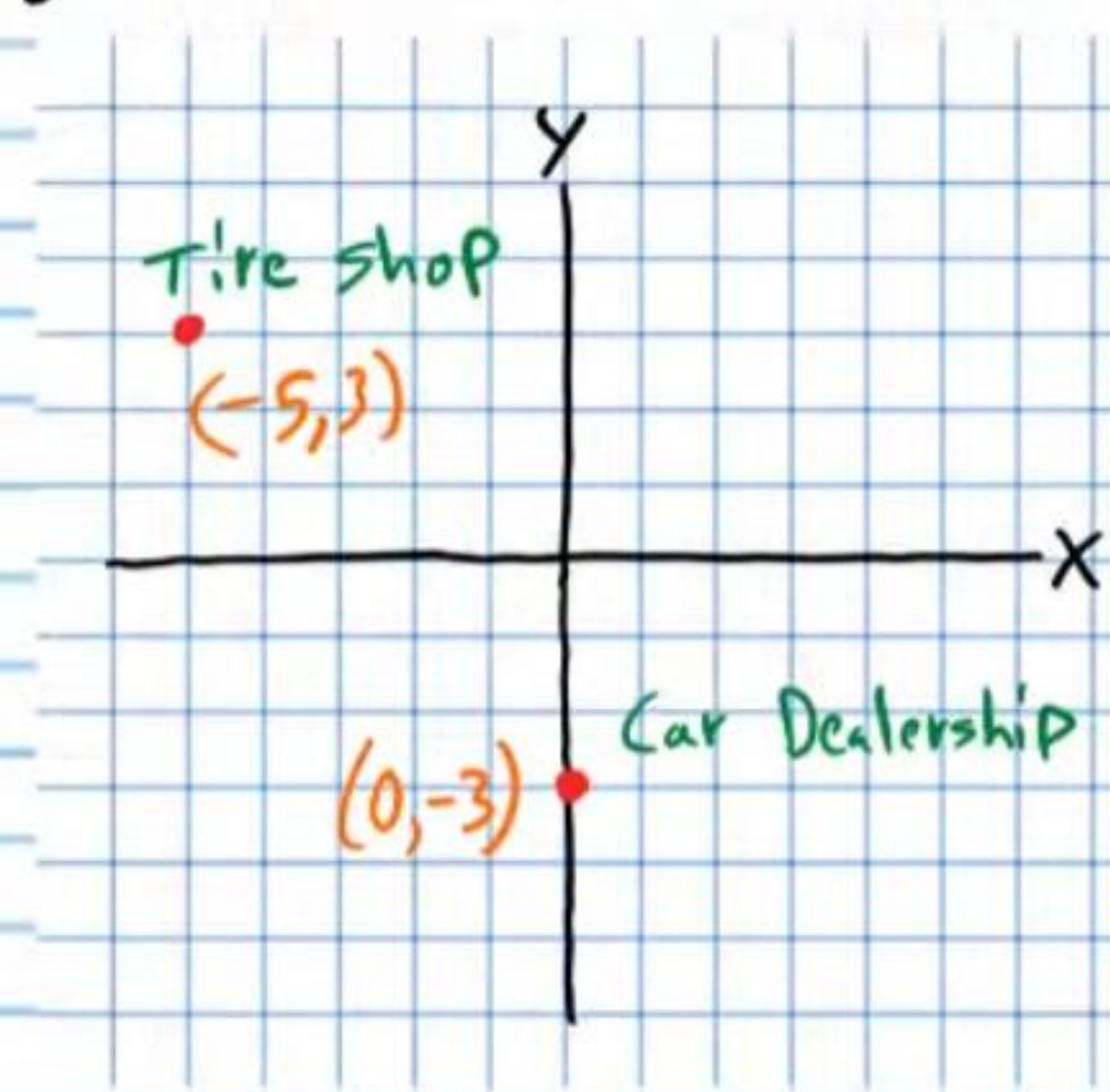
$$\frac{x_1 + x_2}{2} = -5 \quad \left\{ \quad \frac{y_1 + y_2}{2} = 3 \right.$$

$$\frac{0 + x_2}{2} = -5 \quad \left\{ \quad \frac{y_1 + (-3)}{2} = 3 \right.$$

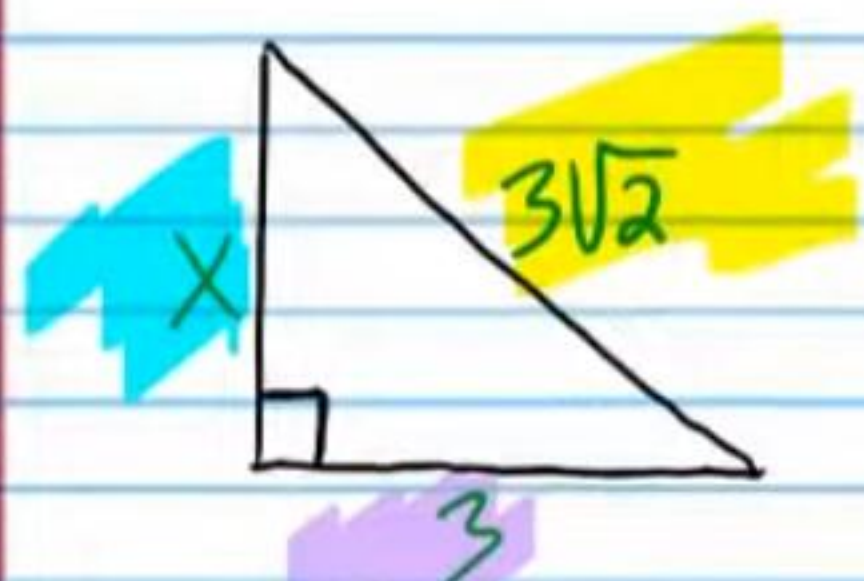
$$\frac{x_2}{2} = -5 \quad \left\{ \quad y_1 + (-3) = 6 \right.$$

$$x_2 = -10 \quad \left\{ \quad y_1 = 9 \right.$$

$$\boxed{(-10, 9)}$$



⑥ Find x .



$$a^2 + b^2 = c^2$$

$$x^2 + 3^2 = (3\sqrt{2})^2$$

$$x^2 + 9 = 3\sqrt{2} \cdot 3\sqrt{2}$$

$$x^2 + 9 = 9 \cdot \sqrt{2} \cdot 2$$

$$x^2 + 9 = 9 \cdot \sqrt{4}$$

$$x^2 + 9 = 9 \cdot 2$$

$$x^2 + 9 = 18$$

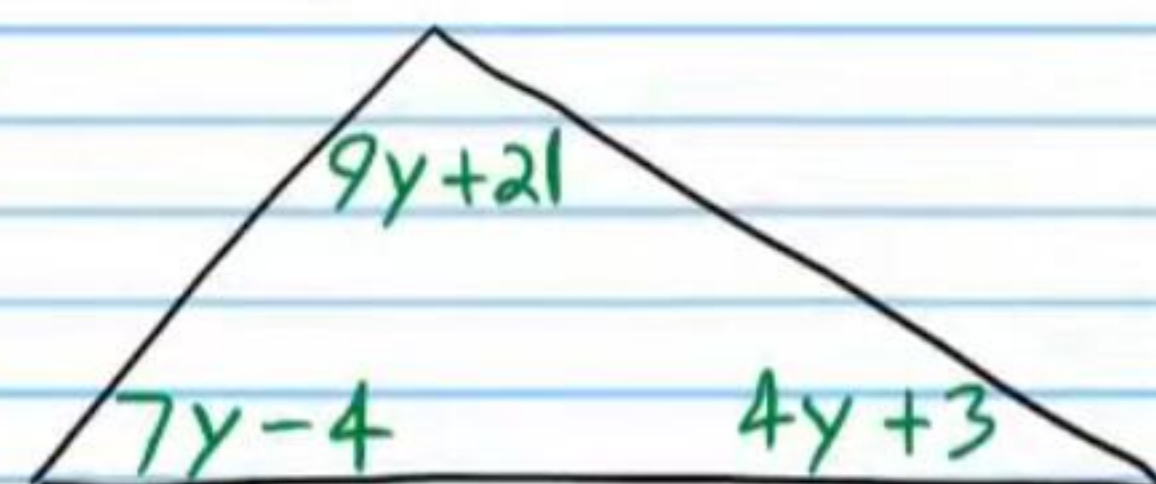
$$x^2 = 9$$

$$\sqrt{x^2} = \pm\sqrt{9}$$

$$x = \pm 3$$

$$\boxed{x = 3}$$

⑦ Find y .



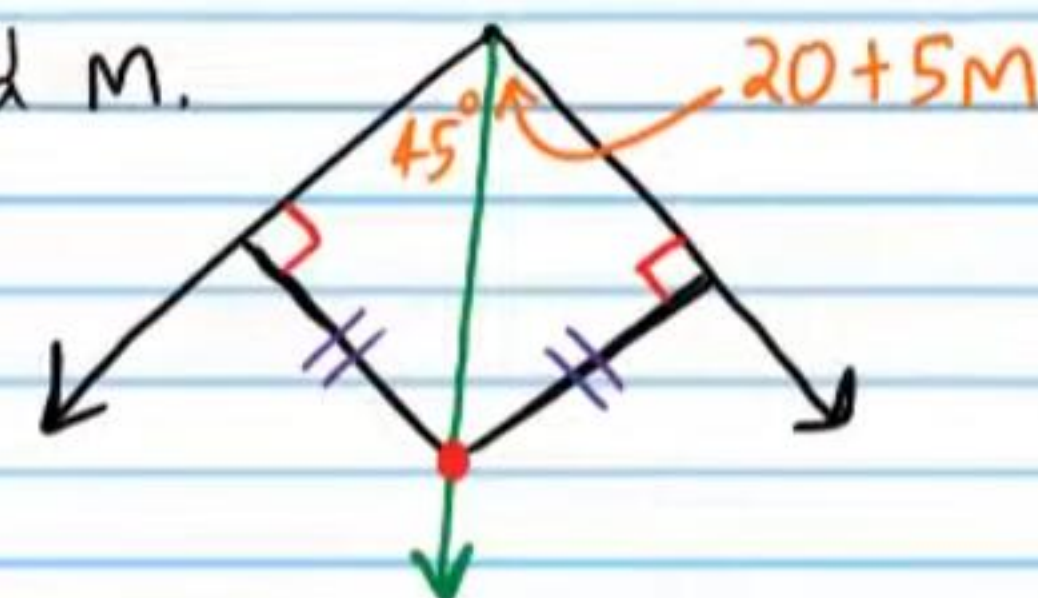
$$9y + 21 + 7y - 4 + 4y + 3 = 180$$

$$20y + 20 = 180$$

$$20y = 160$$

$$\boxed{y = 8}$$

⑧ Find m .

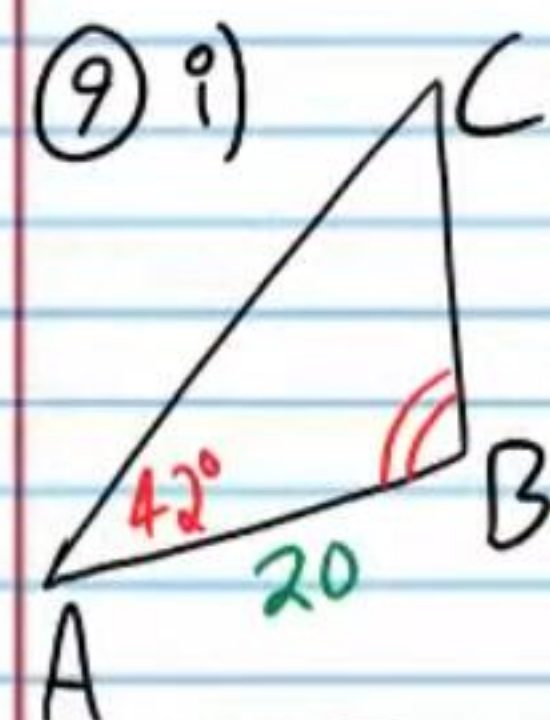


$$45 = 20 + 5m$$

$$25 = 5m$$

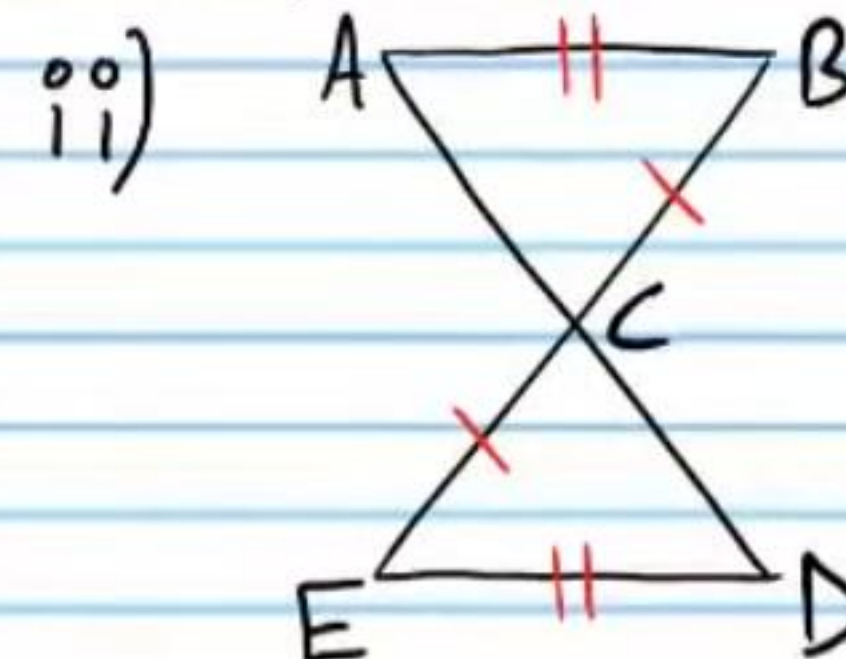
$$\boxed{5 = m}$$

In problem #'s 9-12, determine if the triangles are congruent. If they are congruent, write a proper congruence statement and the triangle postulate or theorem that proves that they are congruent. You do not have to write a formal proof (i.e. a two-column proof).

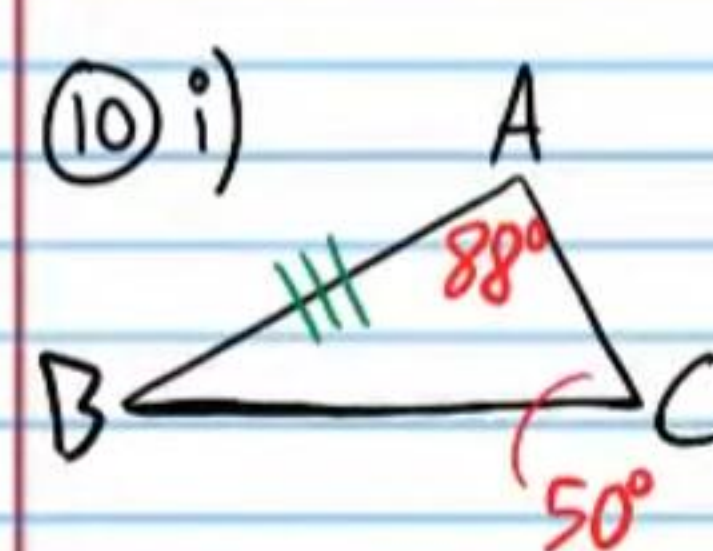


$\boxed{ASA}$

$\boxed{\triangle ABC \cong \triangle DEF}$

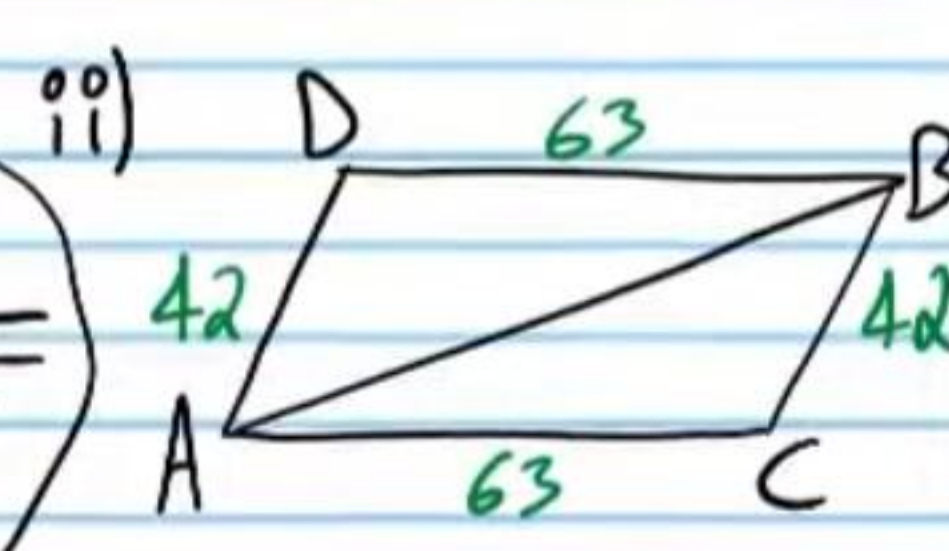


$\boxed{\text{Not Enough Information}}$



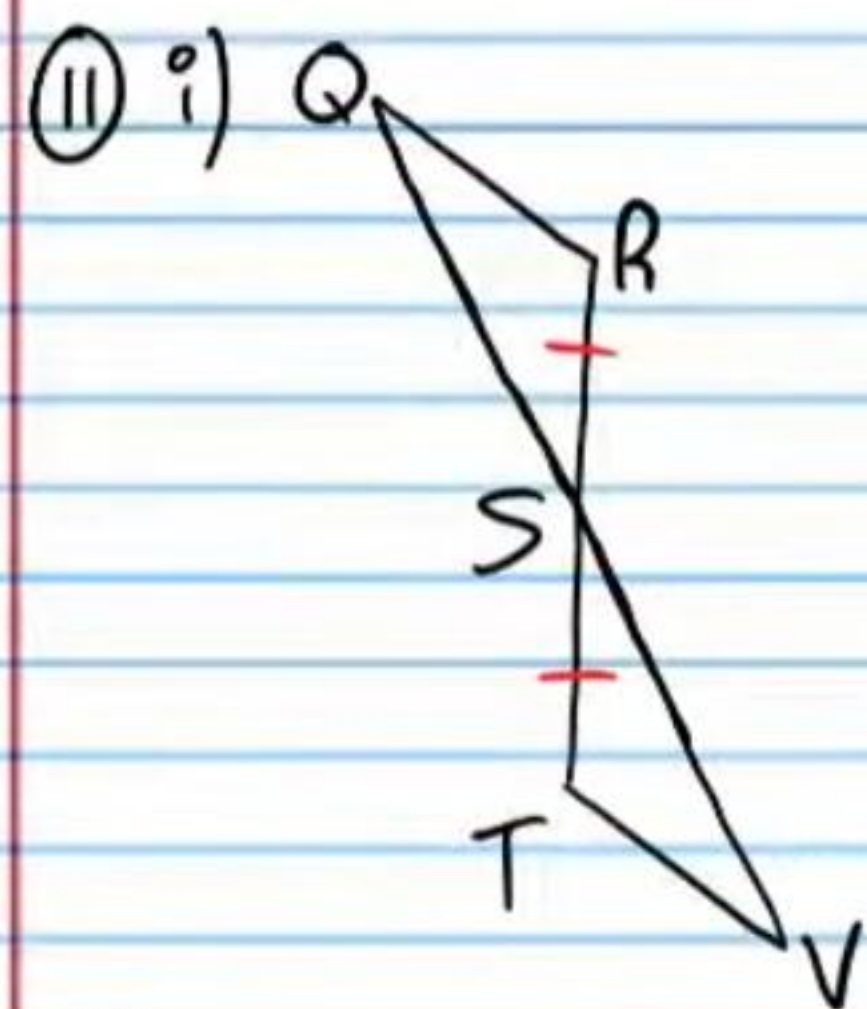
$\boxed{AAS}$

$\boxed{\triangle ABC \cong \triangle DFE}$



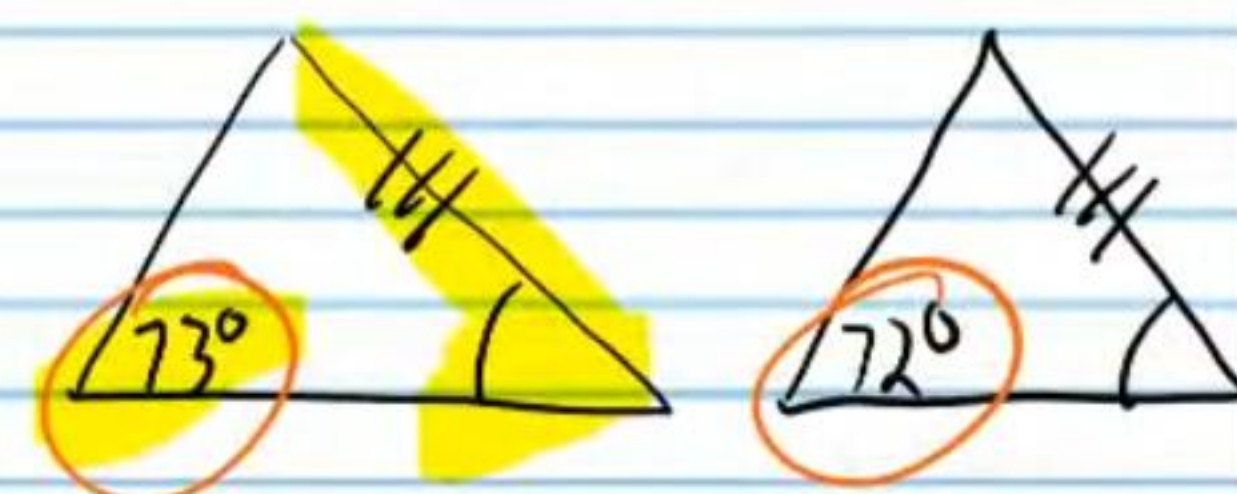
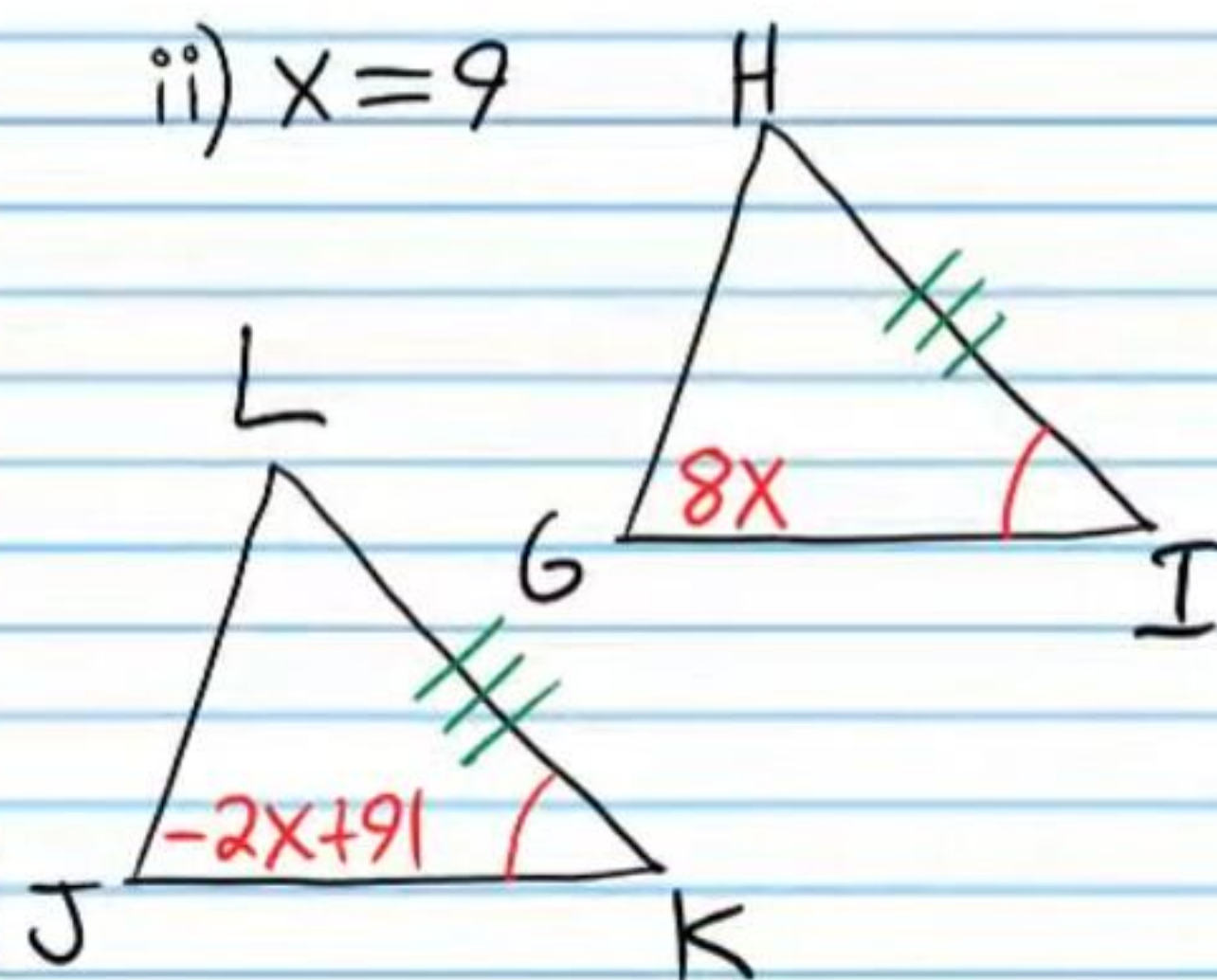
$\boxed{SSS}$

$\boxed{\triangle ABC \cong \triangle BAD}$

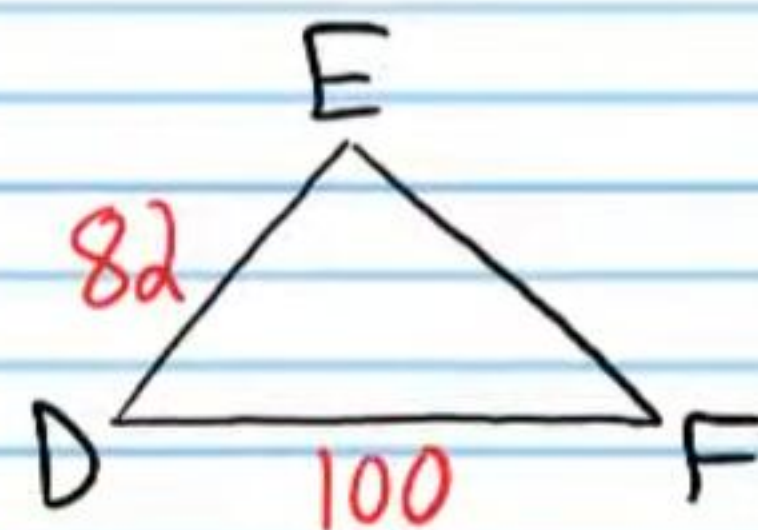
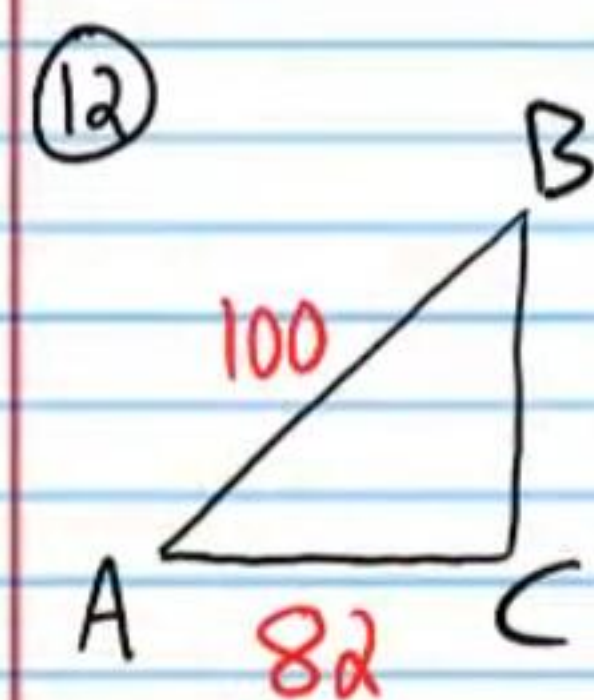


Not Enough Information

ii) $X = 9$

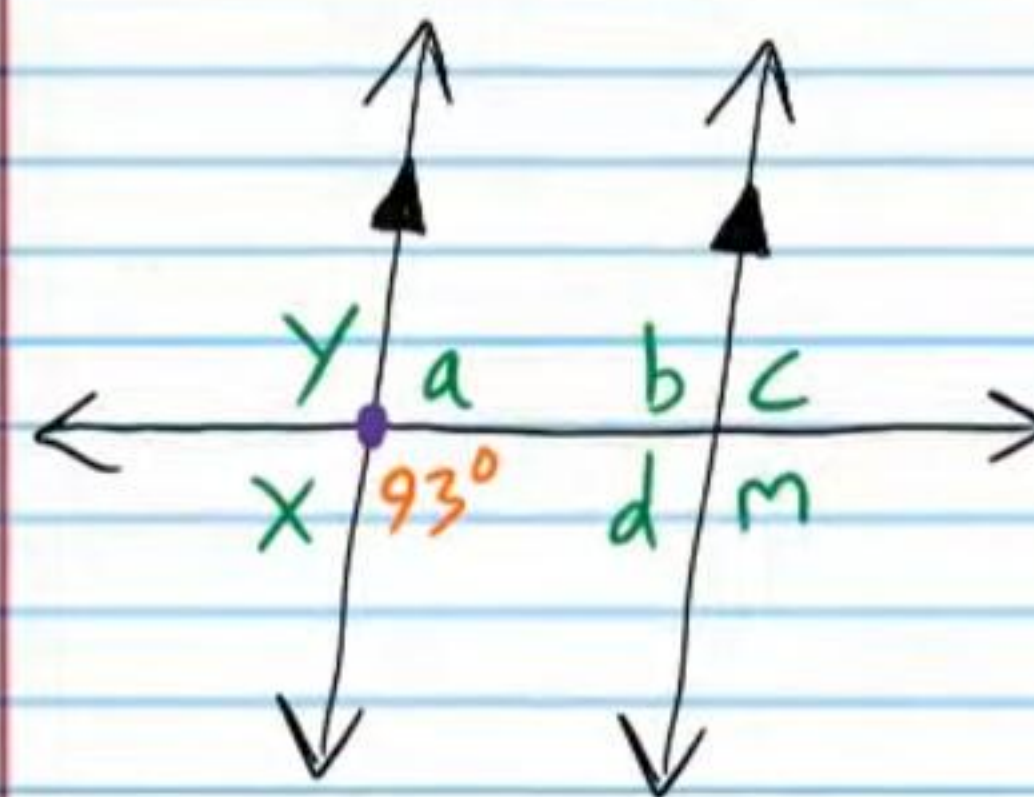


Not Congruent



Not Enough Information

13 Find x, y, a, b, c, d , and m .



$$y = 93^\circ$$

$$c = 87^\circ$$

$$m = 93^\circ$$

$$b = 93^\circ$$

$$93 + a = 180$$

$$a = 87^\circ$$

$$x = 87^\circ$$

$$d = 87^\circ$$

14 Find the distance between the squirrel and the acorn. Each unit on the grid represents 3 meters.

$$(1, 4) \text{ \& } (-2, -1)$$

$$(x_1, y_1) \text{ \& } (x_2, y_2)$$

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

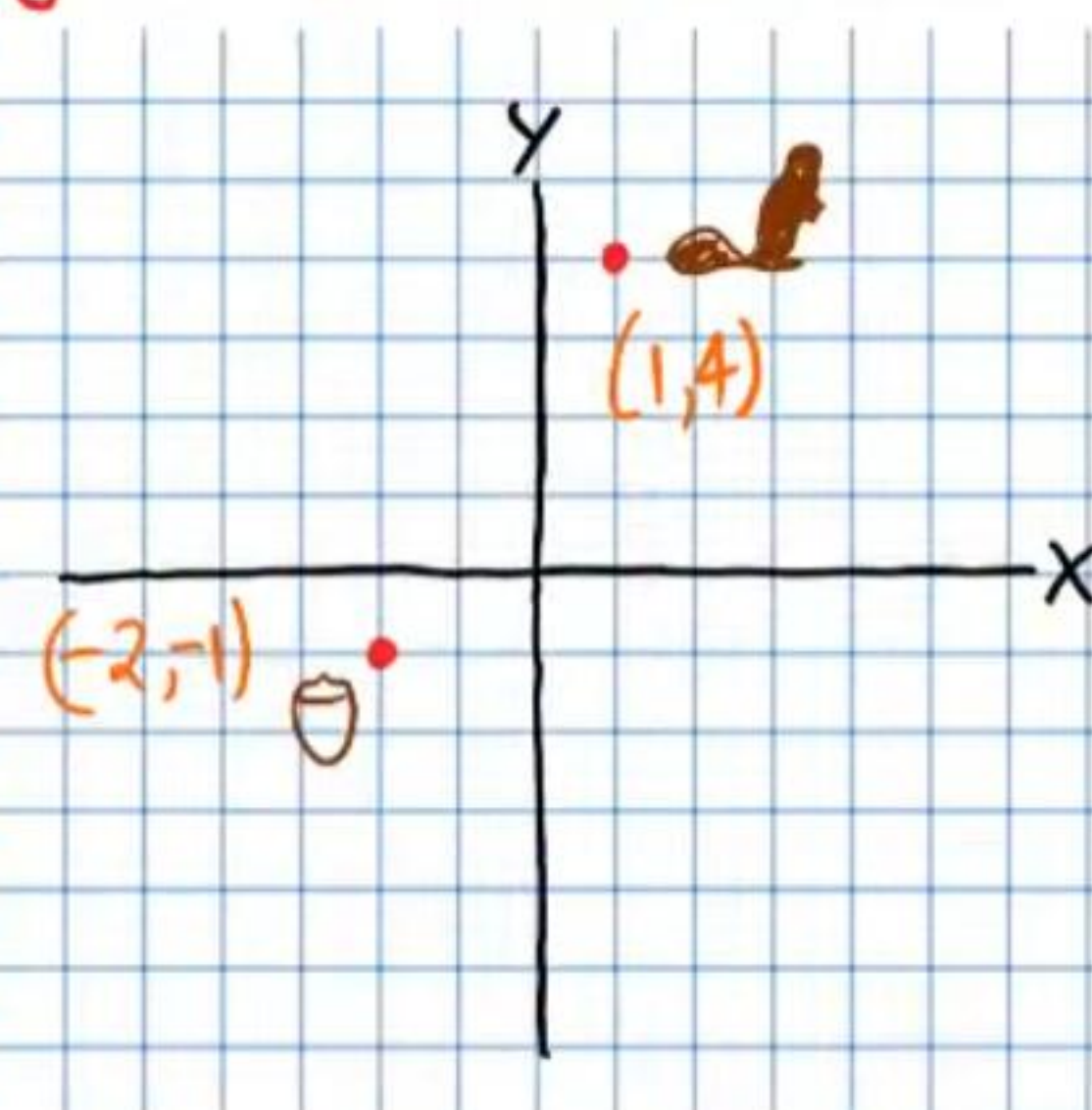
$$D = \sqrt{(-2 - 1)^2 + (-1 - 4)^2}$$

$$D = \sqrt{(-3)^2 + (-5)^2}$$

$$D = \sqrt{9 + 25}$$

$$D = \sqrt{34}$$

$$3\sqrt{34} \text{ m}$$



- ⑮ How much shorter will Casey's path be if he drives on the dirt road rather than the streets?

Street

$$D = 8 + 15$$

$$D = 23 \text{ km}$$

Dirt Road

$$a^2 + b^2 = c^2$$

$$15^2 + 8^2 = c^2$$

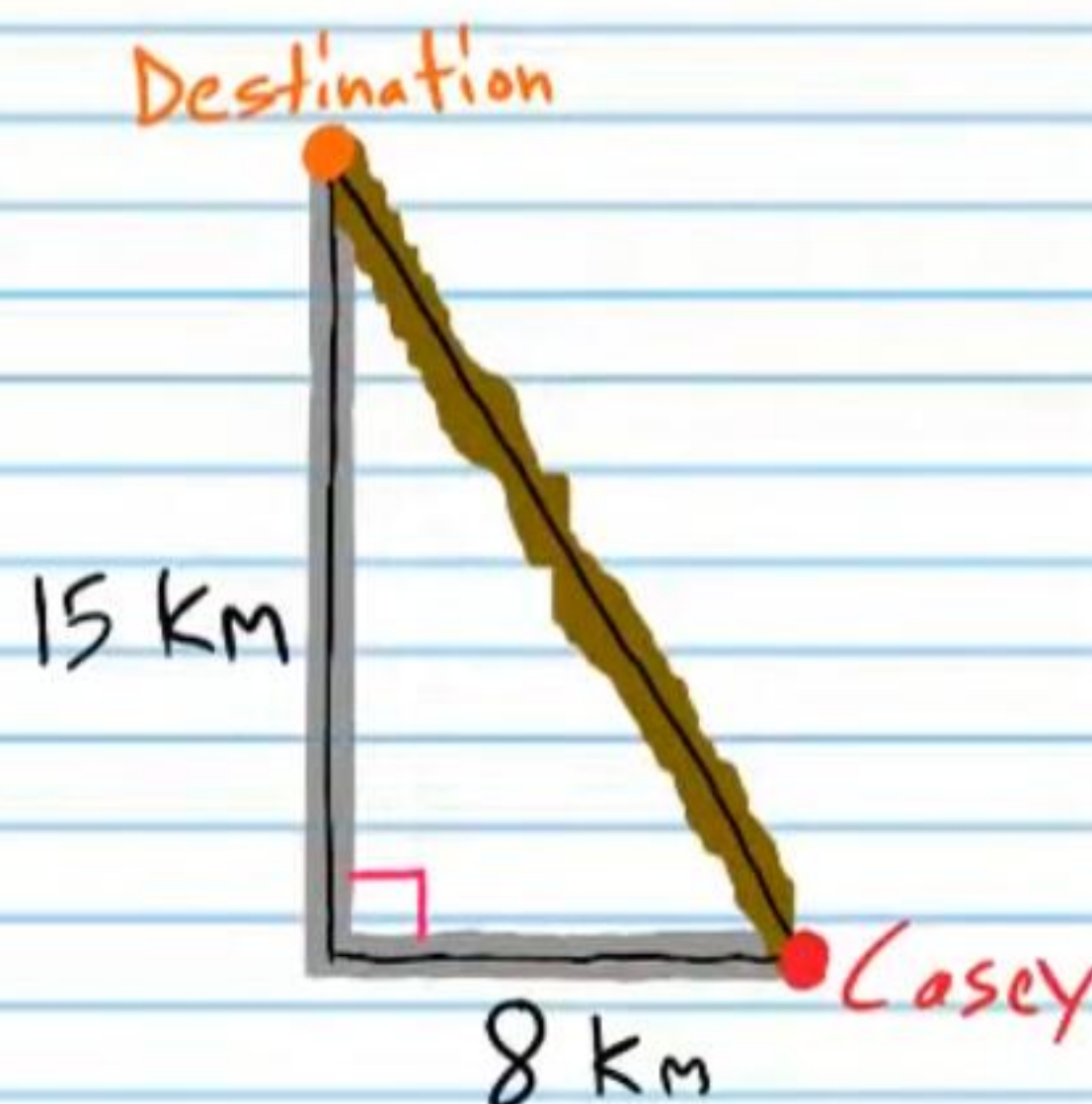
$$225 + 64 = c^2$$

$$289 = c^2$$

$$\pm\sqrt{289} = \sqrt{c^2}$$

$$\pm 17 = c$$

$$17 \text{ km} = c$$



How Much Shorter?

$$23 - 17 = \boxed{6 \text{ km}}$$

- ⑯ Find the midpoint between $(6, -3)$ and $(-3, -2)$.

$$M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right) = M\left(\frac{6 + (-3)}{2}, \frac{-3 + (-2)}{2}\right)$$

$$= \boxed{M\left(\frac{3}{2}, -\frac{5}{2}\right)}$$

- ⑰ If the distance between $(4, 2)$ and $(9, y)$ is $\sqrt{74}$, find the possible values of y .

$$(4, 2) \text{ \& } (9, y)$$

$$(x_1, y_1) \text{ \& } (x_2, y_2)$$

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\sqrt{74} = \sqrt{(9 - 4)^2 + (y - 2)^2}$$

$$\sqrt{74} = \sqrt{5^2 + (y - 2)^2}$$

$$\sqrt{74} = \sqrt{25 + (y - 2)^2}$$

$$(\sqrt{74})^2 = (\sqrt{25 + (y - 2)^2})^2$$

$$74 = 25 + (y - 2)^2$$

$$49 = (y - 2)^2$$

$$\pm\sqrt{49} = \sqrt{(y - 2)^2}$$

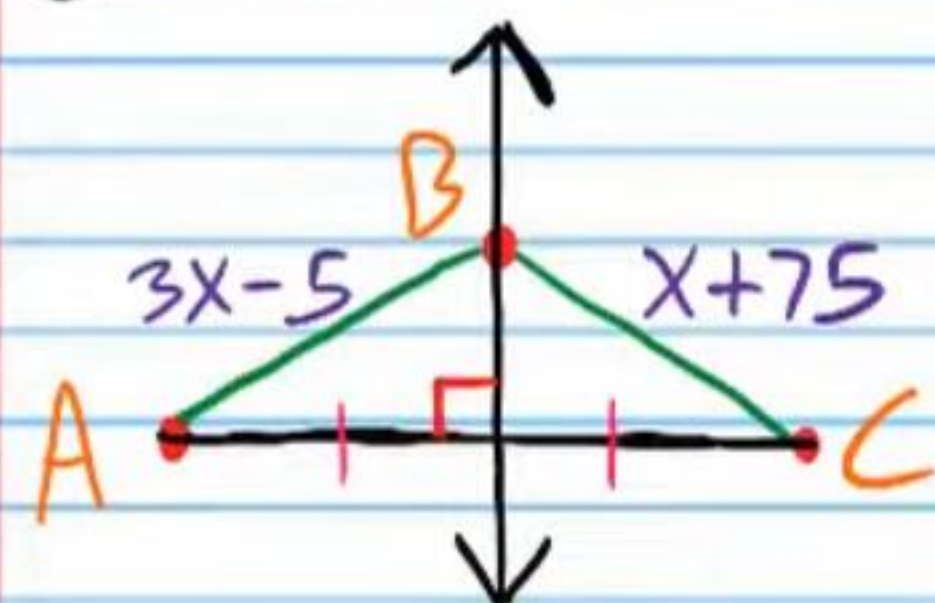
$$\pm 7 = y - 2$$

$$\pm 7 = y - 2$$

$$7 = y - 2 \text{ or } -7 = y - 2$$

$$\boxed{9 = y} \text{ or } \boxed{-5 = y}$$

⑮ Find BC.



$$3x-5 = x+75$$

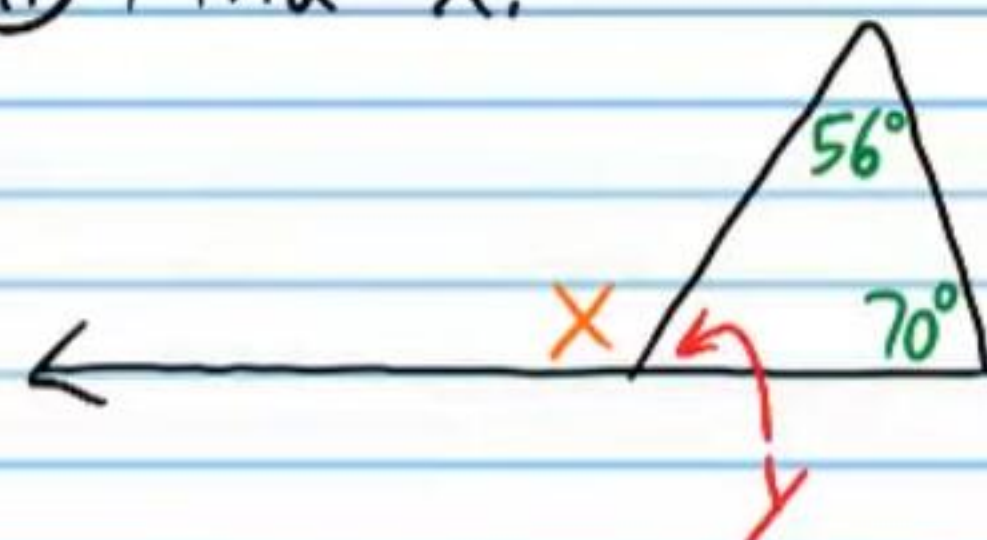
$$2x = 80$$

$$x = 40$$

$$BC = 40 + 75$$

$$\boxed{BC = 115}$$

⑯ Find X.



$$56 + 70 + y = 180$$

$$126 + y = 180$$

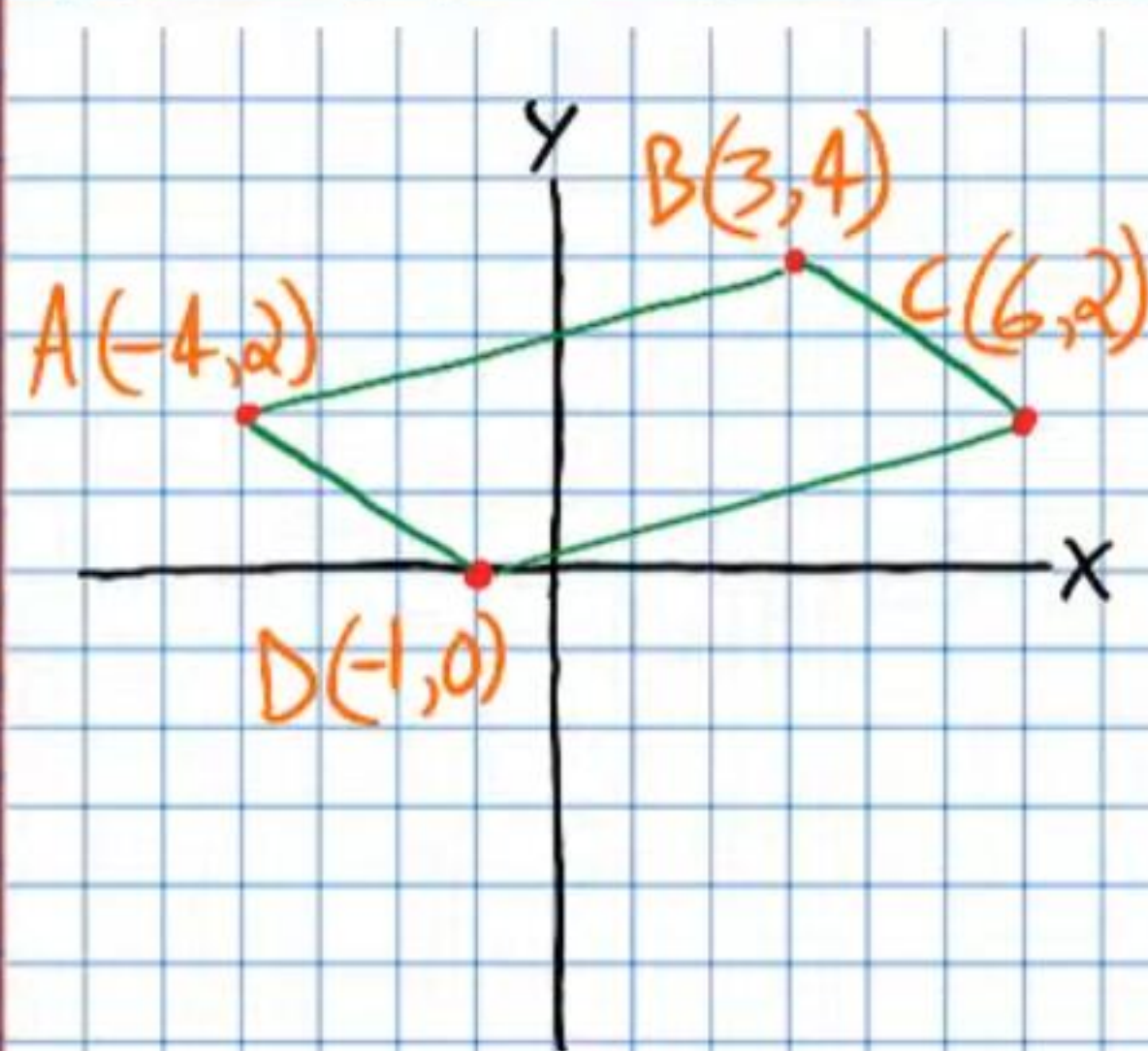
$$y = 54^\circ$$

$$x + y = 180$$

$$x + 54 = 180$$

$$\boxed{x = 126^\circ}$$

⑰ Prove that the diagonals of the quadrilateral below bisect each other.



$$M_{BD}\left(\frac{3+(-1)}{2}, \frac{4+0}{2}\right) = M_{BD}\left(\frac{2}{2}, \frac{4}{2}\right)$$

$$= M_{BD}(1, 2)$$

$$M_{AC}\left(\frac{-4+6}{2}, \frac{2+2}{2}\right) = M_{AC}\left(\frac{2}{2}, \frac{4}{2}\right)$$

$$= M_{AC}(1, 2)$$

$$M_{BD}(1, 2) = M_{AC}(1, 2)$$

Therefore, $\overline{AC}$ & $\overline{BD}$ bisect each other.

⑱ Show that the triangle below is isosceles.

$$D_{AB} = \sqrt{(-5 - (-2))^2 + (-3 - 2)^2}$$

$$D_{AB} = \sqrt{(-5 + 2)^2 + (-5)^2}$$

$$D_{AB} = \sqrt{(-3)^2 + (-5)^2}$$

$$D_{AB} = \sqrt{9 + 25}$$

$$D_{AB} = \sqrt{34}$$

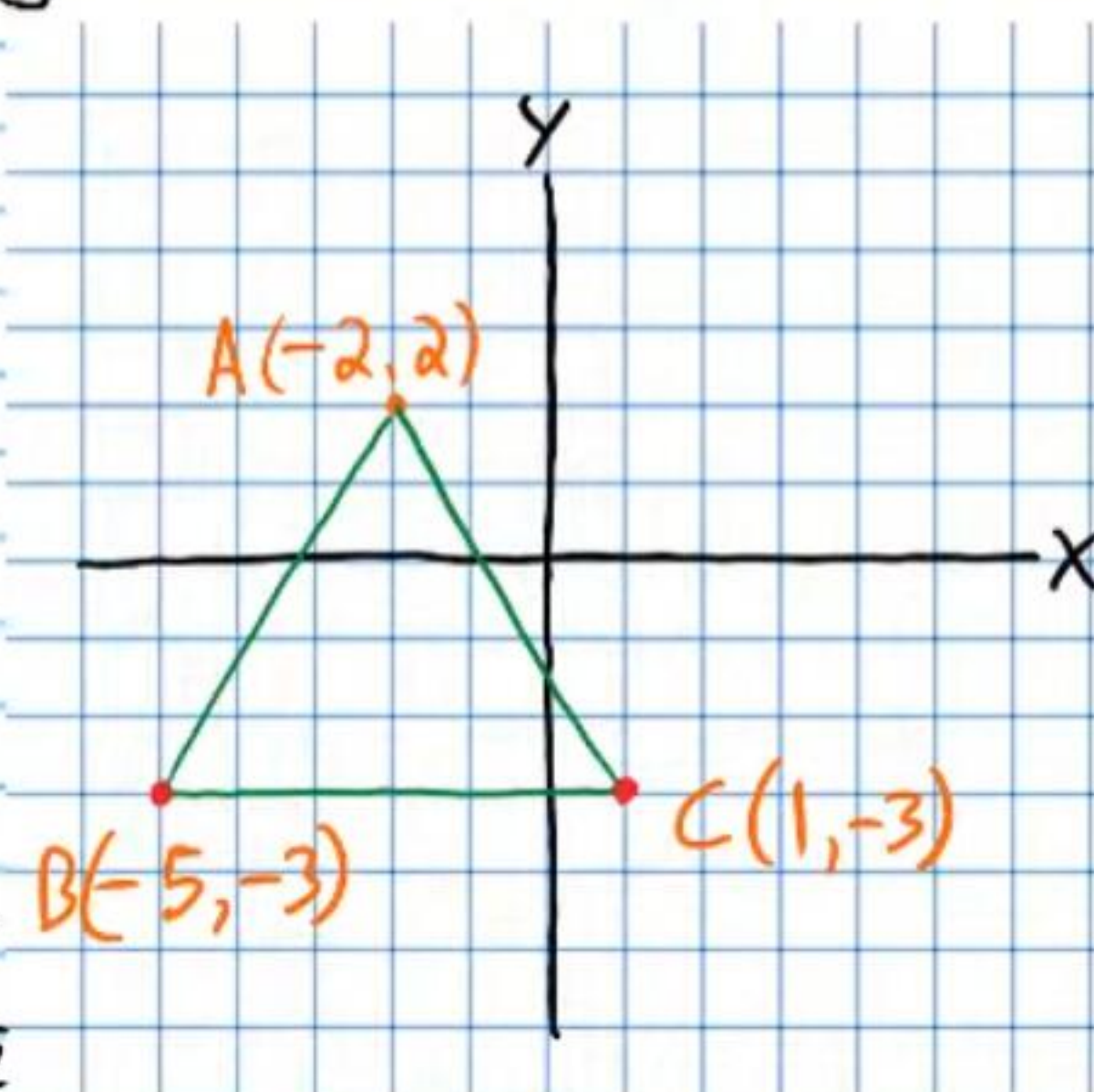
$$D_{AC} = \sqrt{(1 - (-2))^2 + (-3 - 2)^2}$$

$$D_{AC} = \sqrt{(1 + 2)^2 + (-5)^2}$$

$$D_{AC} = \sqrt{3^2 + 25}$$

$$D_{AC} = \sqrt{9 + 25}$$

$$D_{AC} = \sqrt{34}$$

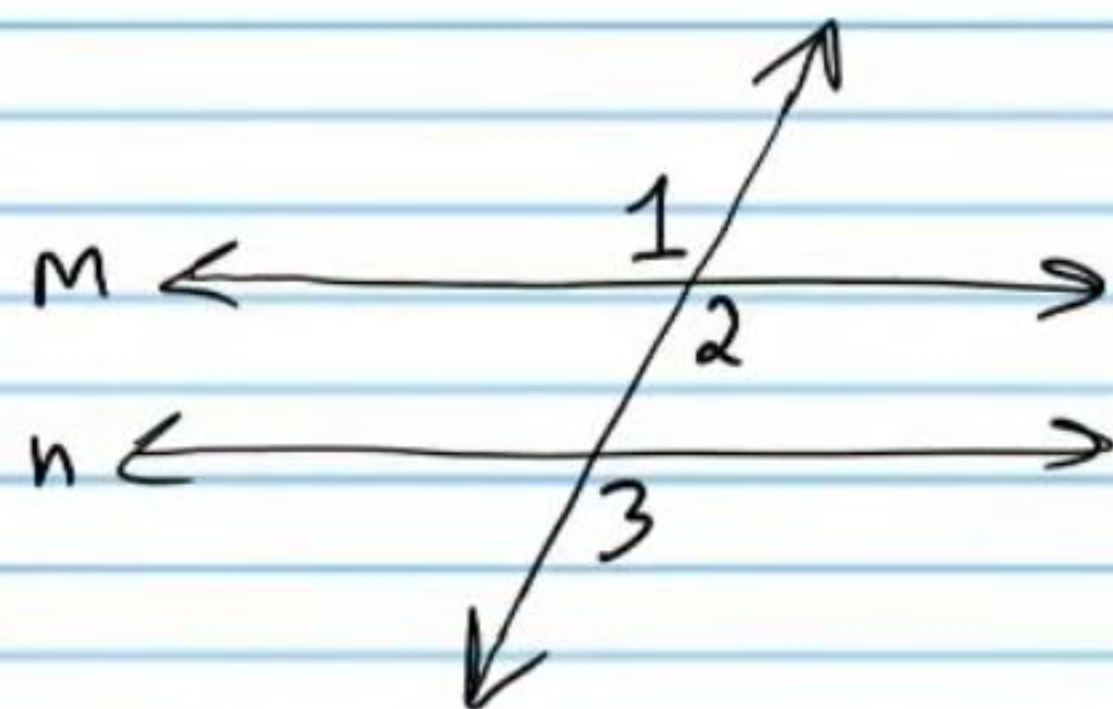


$$D_{AB} = D_{AC}$$

Therefore, the triangle is isosceles.

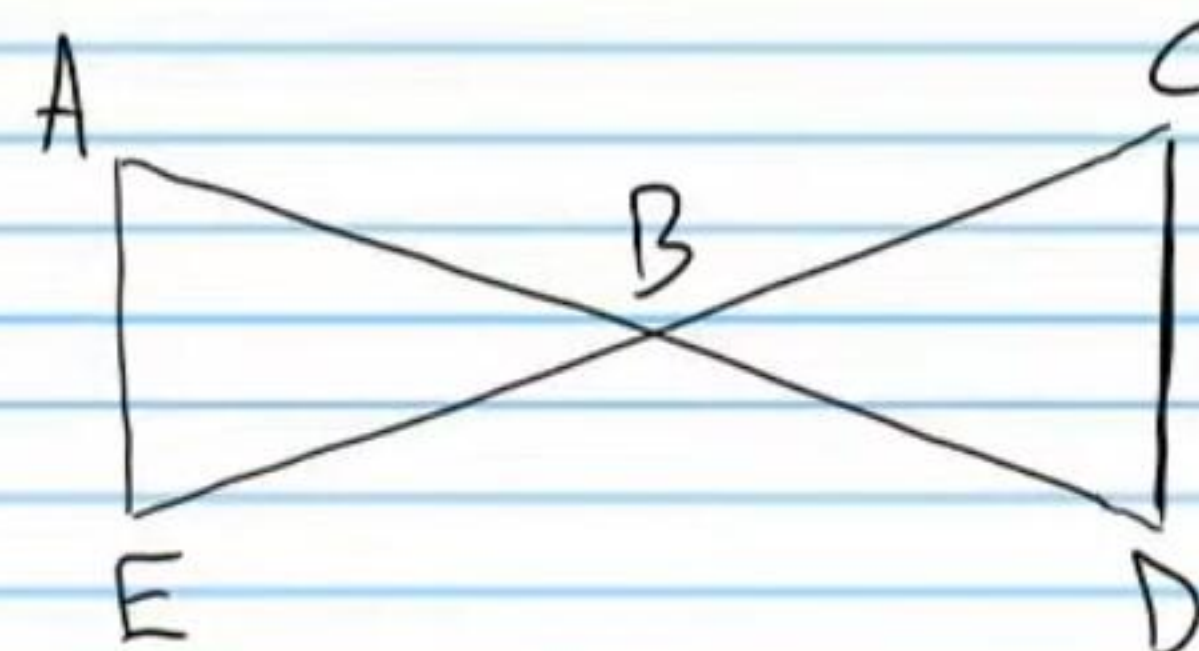
Show that the following statements are true using two-column proofs. You might not use the same number of steps written in Roman numerals. You may use more or less.

② If $\angle 1 \cong \angle 3$, then $m \parallel n$. Note: This is the proof of the Alternate Exterior Angles Converse Theorem. Use the Corresponding Angles Converse Postulate and the Vertical Angles Theorem.



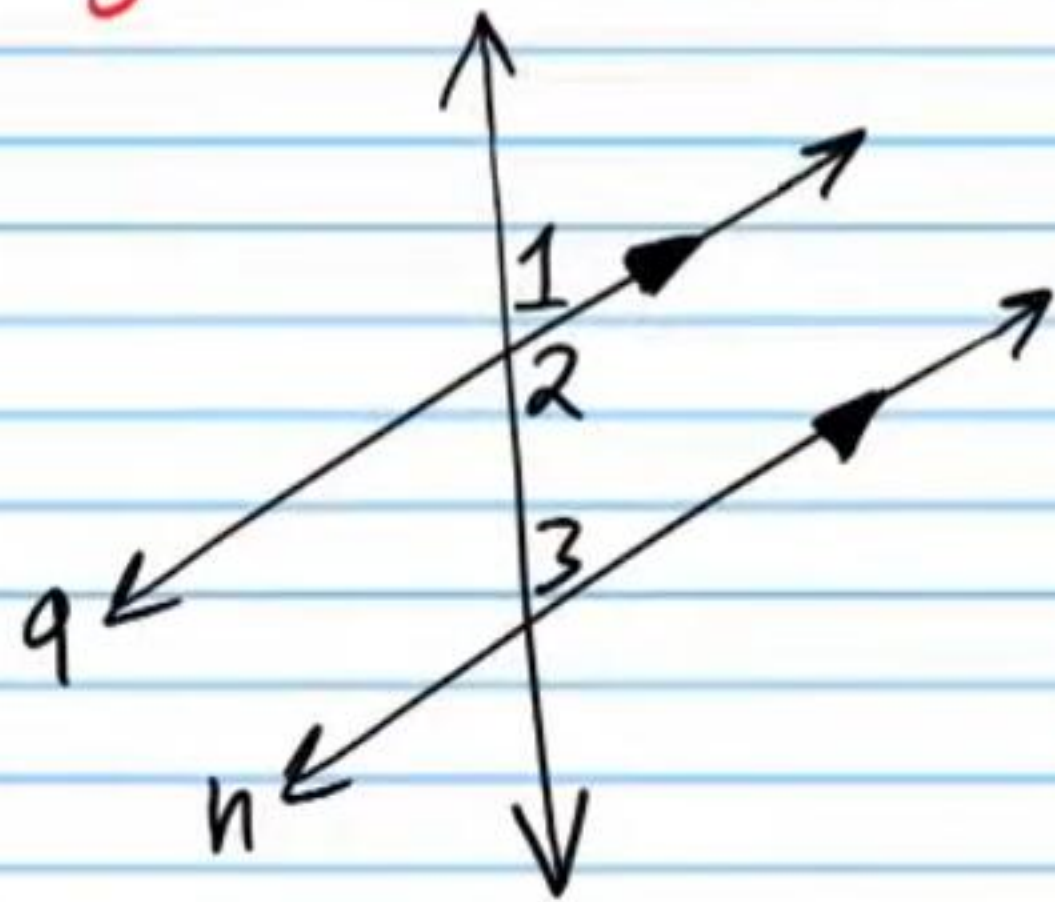
Statement	Reason
I) $\angle 1 \cong \angle 3$	Given
II) $\angle 1 \cong \angle 2$	Vert. $\angle$'s Thm.
III) $\angle 3 \cong \angle 2$	Trans. Prop of $\cong$.
IV) $m \parallel n$	CACP

② If $\angle E \cong \angle C$ and B is the midpoint of $\overline{AD}$, then $\overline{EB} \cong \overline{BC}$.



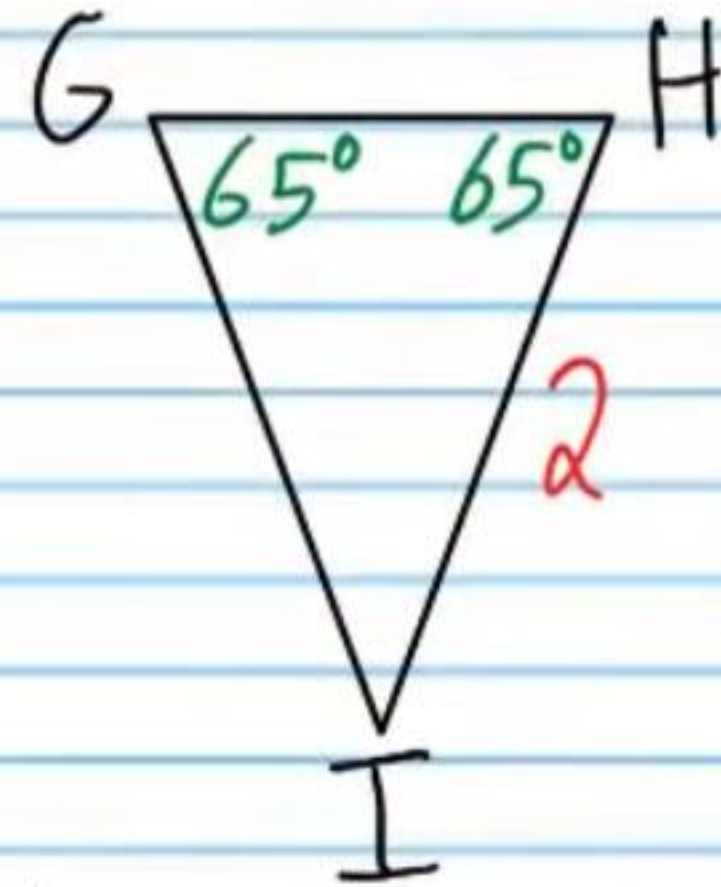
Statement	Reason
I) $\angle E \cong \angle C$	Given.
II) B is the midpoint of $\overline{AD}$	Given.
III) $\overline{AB} \cong \overline{BD}$	Def. of midpoint.
IV) $\angle ABE \cong \angle CBD$	Vert. $\angle$'s Thm
V) $\triangle ABE \cong \triangle CBD$	AAS
VI) $\overline{EB} \cong \overline{BC}$	CPCTC

②④ If $q \parallel n$ below, then $\angle 2$ and $\angle 3$ are supplementary. Note: This is the proof of the Consecutive Interior Angles Theorem. *You must use the Corresponding Angles Postulate and the Linear Pair Postulate.*



Statement	Reason
I) $q \parallel n$	Given.
II) $\angle 1 \cong \angle 3$	CAP
III) $\angle 1$ & $\angle 2$ form a linear pair.	Given (diagram)
IV) $\angle 1$ & $\angle 2$ are supplementary.	LPP
V) $m\angle 1 + m\angle 2 = 180^\circ$	Def. of supp $\angle$'s
VI) $m\angle 1 = m\angle 3$	Def. of $\cong \angle$'s.
VII) $m\angle 3 + m\angle 2 = 180$	Subs. Prop of =
VIII) $\angle 3$ & $\angle 2$ are supplementary	Def. of supp $\angle$'s.

②⑤ If $m\angle G = 65^\circ$, $m\angle H = 65^\circ$, and $HI = 2$, then $GI = 2$. Hint: Use the Isosceles Triangle Converse Theorem.



Statement	Reason
I) $m\angle G = 65^\circ$ & $m\angle H = 65^\circ$	Given.
II) $m\angle G = m\angle H$	Trans. Prop of =.
III) $\angle G \cong \angle H$	Def. of $\cong \angle$'s.
IV) $\overline{GI} \cong \overline{HI}$	Isos Δ Conv Thm
V) $GI = HI$	Def. of $\cong$ seg.
VI) $HI = 2$	Given
VII) $GI = 2$	Trans. Prop of =.